

Twitter Sentiment Analysis: DistilBERT vs Traditional Models

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Problem Statement

Twitter posts on various subjects

Want to know writer's opinion about a **specific subject**

- Useful for market research

Basic sentiment analysis is a common pretrained task for LLMs

- Out of the box pipeline for HuggingFace
- Only finds **general** sentiment independent of subject
- Doesn't identify irrelevant data

Problem Statement

Comparing Large Language Model vs traditional models

- LLMs are expensive to train and fine-tune
- Want to know if they are significantly more accurate

Dataset

Sourced from Kaggle

Tweet content and labeled sentiment towards provided entity

- Means of sentiment assignment is not stated (crowdsourcing, study, etc.)
- Train: 74682 rows
- Test: 1758 rows
- Format: Tweet ID | Entity | Sentiment | Tweet content

Artificial permutations of tweet contents in training data

- 6 per tweet

Dataset

Three sentiments

- Positive, Neutral, Negative
- Additional label for Irrelevant tweets

32 Entities

- Mostly video games (Borderlands, CS-GO)
- Some companies (Verizon, Amazon, Home Depot)

Tweet Id	352
Entity	Amazon
Sentiment	Neutral
Tweet Content	BBC News - Amazon boss Jeff Bezos rejects clai...

2410	Borderlands	Negative	why does like every man in borderlands have slicked back hair haven't you heard of bangs you stupid ...
2410	Borderlands	Negative	Why, like every man in border countries, have you not heard of bangs, you stupid assholes?
2410	Borderlands	Negative	Why, like everyone else in the border countries, have your hair fallen off, have you not heard that ...
2410	Borderlands	Negative	why does like<unk> man in borderlands have slicked back hair or you heard of bangs you stupid asshol...
2410	Borderlands	Negative	why Beth does like every man in borderlands have she slicked that back hair haven ' t you heard enou...
2410	Borderlands	Negative	why does practically every man in France have slicked back hair haven't you heard of bangs the greed...

Preprocessing

Removed any data points with missing values

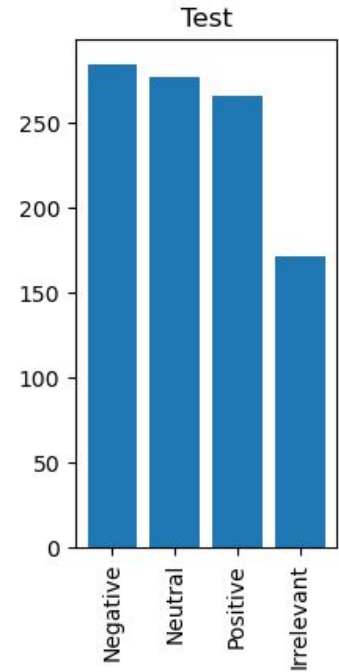
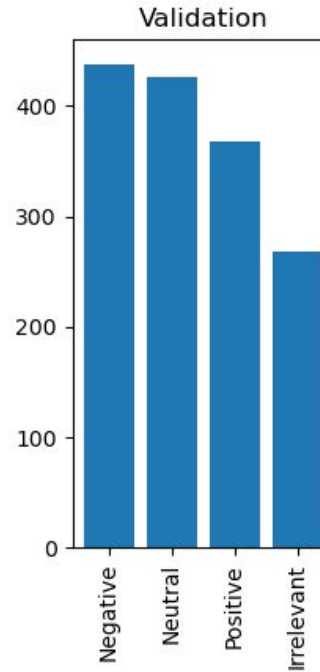
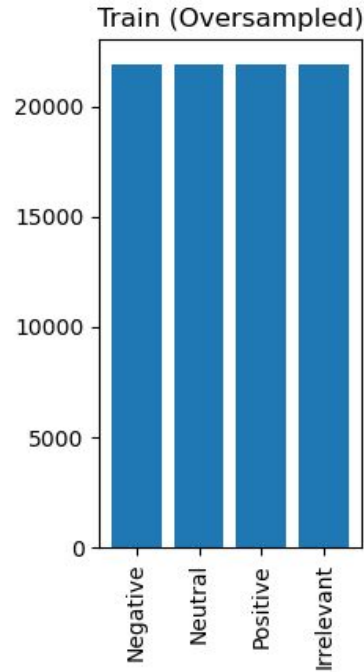
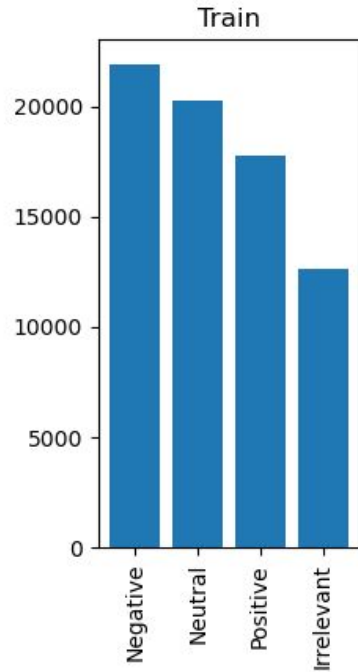
- 686 rows removed in training set

Separated 1500 rows into validation set using `scipy train_test_split`

Balanced classes in training set using oversampling

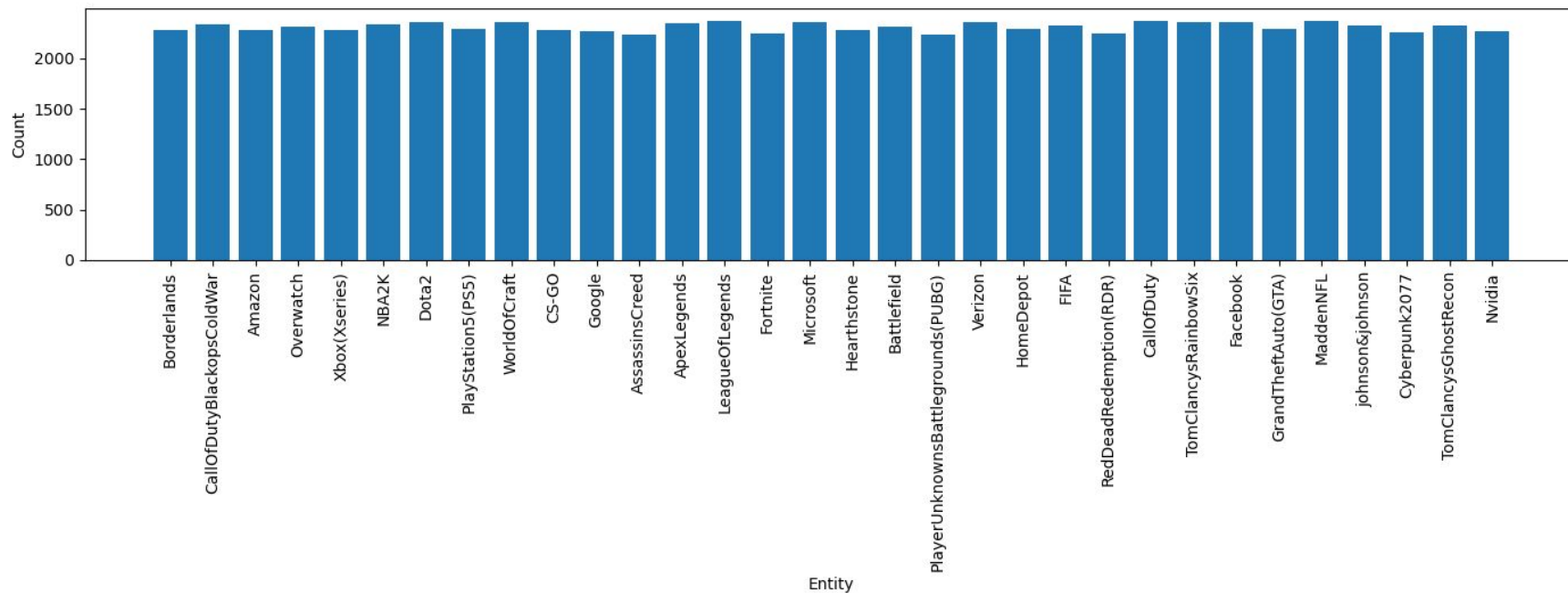
- More negative tweets than neutral or positive
- Much fewer irrelevant tweets

Preprocessing - Balancing



Preprocessing - Balancing

Reasonably well balanced across entities



TF-IDF

Document Frequency between 0.02 and 0.6

- 998 features
- Ensures features are not overly general or specific

Entities one-hot encoded

- Concatenated to start of TF-IDF matrix
- 32 entities

1030 total features



Models

Basic

- Linear Support Vector Machine (SVM)
- K Nearest Neighbors (KNN)

Ensembles

- Random Forest
- AdaBoost

Large Language Model

- DistilBERT

Basic Models

Linear SVM

Used LinearSVC from sklearn.svm

Tuned using GridSearchCV on training set

- Scored by MCC
- Grid:
 - loss : hinge, squared_hinge
 - dual : True, False
 - C : 0.01, 0.1, 0.5, 1.0
 - max_iter : 500, 1000, 2000, 5000, 10000, 20000

Best model used **C = 1.0**, **Lagrangian dual** optimization, **hinge** loss, **500 iterations**

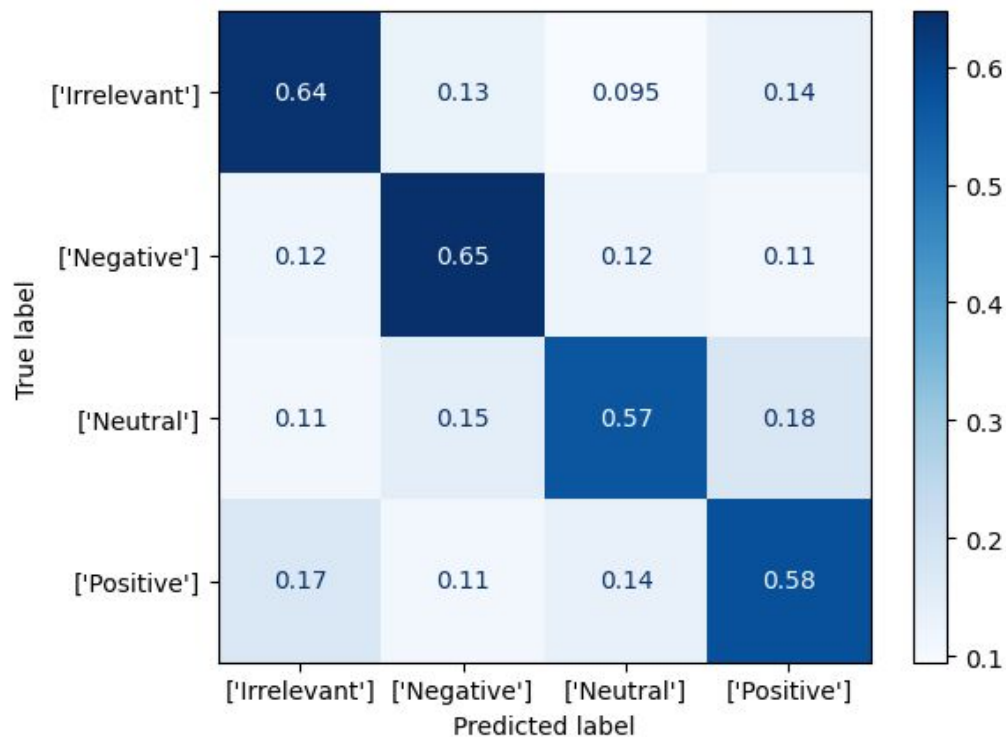
Results - Linear SVM

Test Set

MCC	0.558
Avg. Precision	0.663
Avg. Recall	0.669
Avg. F1	0.664
Accuracy	0.670

Train Set

MCC	0.513
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KNN

Used KNeighborsClassifier from sklearn.neighbors

Tuned using GridSearchCV on training set

- Scored by MCC
- Grid:
 - n_neighbors : 3, 5, 7, 9, 11, 13
 - weights : uniform, distance

Best model used **5 neighbors** and **uniform** weighting

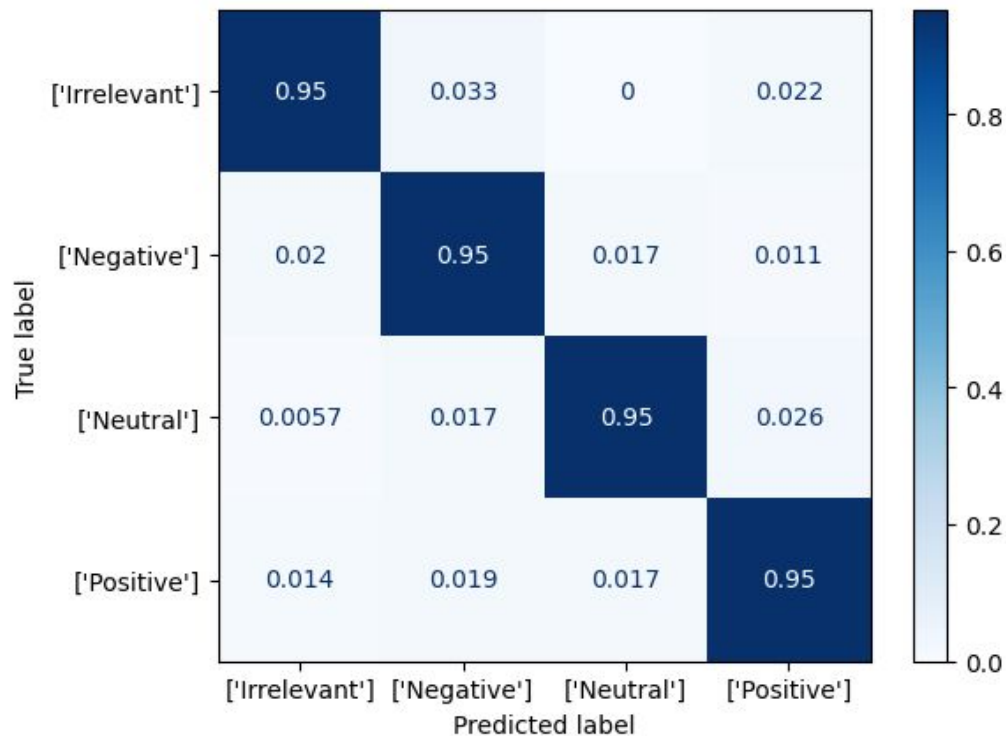
Results - KNN

Test Set

MCC	0.954
Avg. Precision	0.963
Avg. Recall	0.966
Avg. F1	0.965
Accuracy	0.966

Train Set

MCC	0.948
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Ensemble Models

Random Forest

Used RandomForestClassifier from sklearn.ensemble

Tuned using GridSearchCV on training set

- Scored by MCC
- Grid:
 - n_estimators : 10, 20, 50, 100, 200
 - min_samples_split : 2, 3, 4

Best model had **100 estimators** and minimum split of **4**

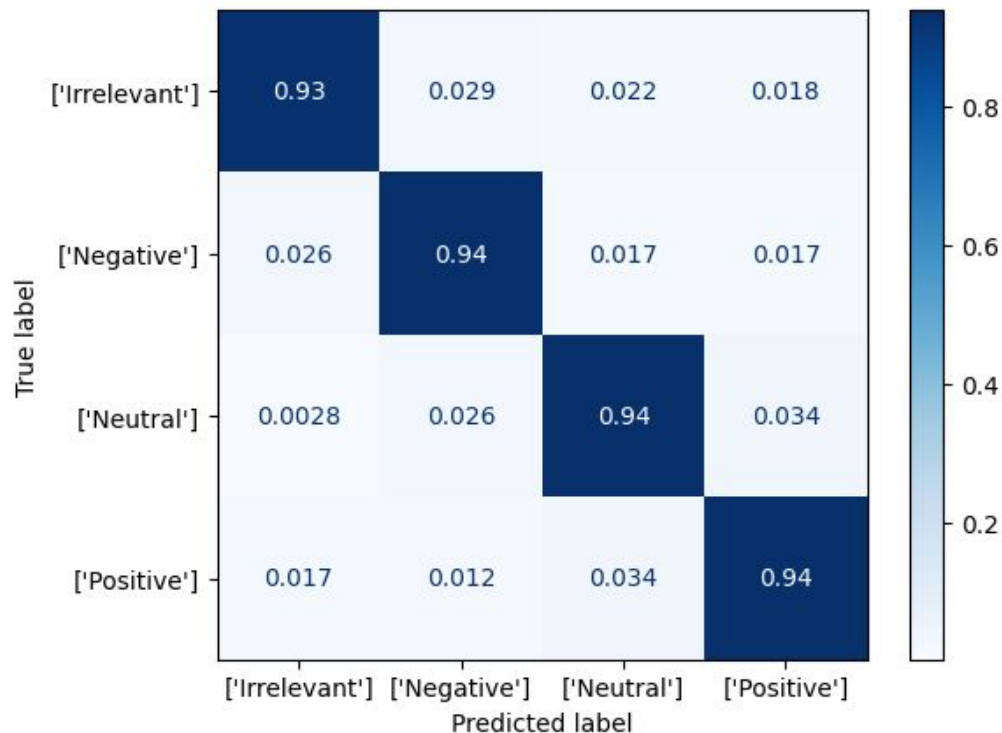
Results - Random Forest

Test Set

MCC	0.935
Avg. Precision	0.954
Avg. Recall	0.949
Avg. F1	0.951
Accuracy	0.952

Train Set

MCC	0.975
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AdaBoost

Used AdaBoostClassifier from sklearn.ensemble

- Default configuration - Decision Trees (max depth of 1)

Tuned using GridSearchCV on training set

- Scored by MCC
- Grid:
 - n_estimators : 100, 200, 500
 - learning_rate : 0.7, 0.9, 1.0

Best model had **500 estimators** and learning rate of **1.0**

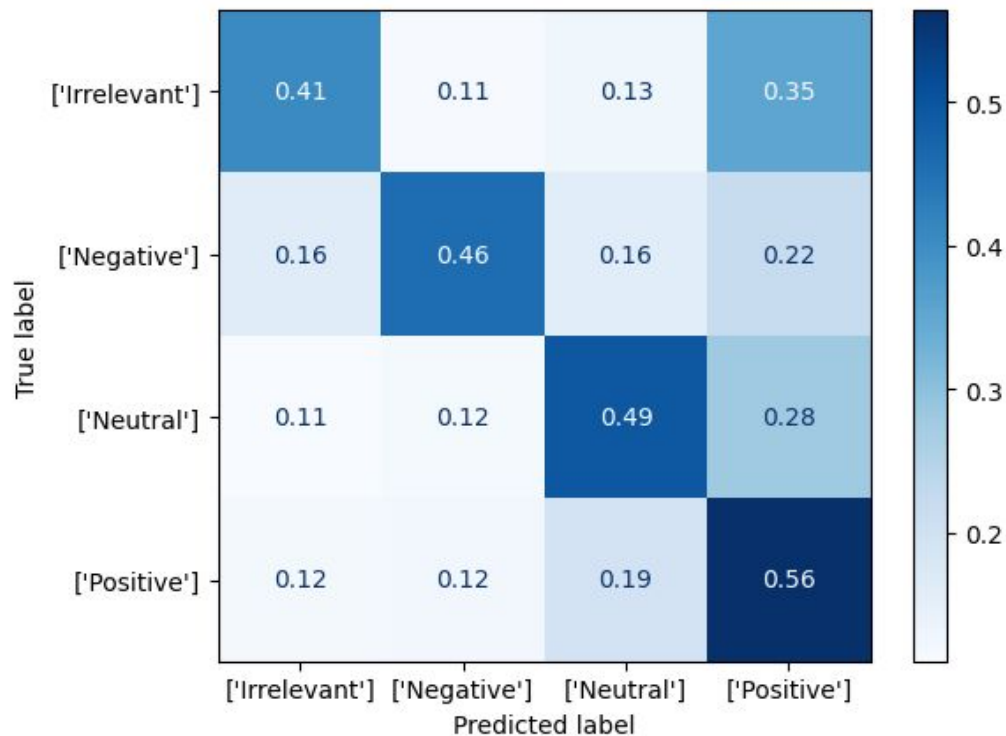
Results - AdaBoost

Test Set

MCC	0.319
Avg. Precision	0.491
Avg. Recall	0.481
Avg. F1	0.482
Accuracy	0.495

Train Set

MCC	0.307
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DistilBERT

What is BERT?

Bidirectional Encoder Representations from Transformers

Published by Google's AI Language group

- Encoder-only model
 - All to all attention
- Can “see” words both before and after current token
- Most models either left-to-right or right-to-left
- Trained using masked input (fill in the blanks)
 - Otherwise could “see” current token - trivial
- **Same general architecture for pretraining and fine-tuning**

BERT-base has 12 layers, hidden size of 768, and 12 self-attention heads [1]

DistilBERT

“Distilled” model

- Same general architecture as BERT
- Half as many layers
- Initialized from BERT model

Trained as a “student” with original BERT as “teacher”

- Retains 97% of BERT performance
- Significantly faster computation [2]

Fine Tuning DistilBERT

Tokenized data has higher dimensionality

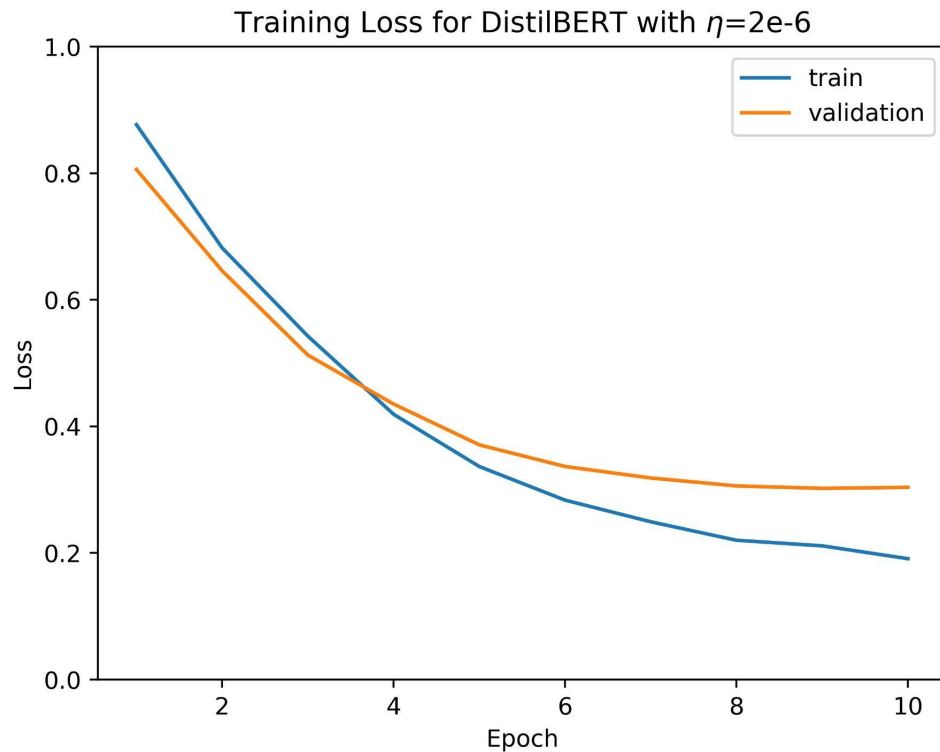
- Training: 160028 rows
- Entity label to start of tweet content

Fine tuned using HuggingFace trainer

- Run on Google Colab
 - NVIDIA L4 GPU
 - Roughly 30 minutes per epoch
- Learning rates of 5e-5, 5e-6, 2e-6, and 5e-7
 - Best was 2e-6 after 9 epochs (validation loss of 0.302)
 - Otherwise default parameters
- Accuracy metric
- Combination of logit and binary cross entropy loss

```
training_args = TrainingArguments(  
    output_dir="drive/MyDrive/BERT/%s"%run_name,  
    eval_strategy="epoch",  
    save_strategy="epoch",  
    load_best_model_at_end=True,  
    learning_rate=2e-6,  
    per_device_train_batch_size=16,  
    per_device_eval_batch_size=16,  
    num_train_epochs=10,  
    weight_decay=0.01,  
    push_to_hub=False,  
    report_to="none"  
)
```

Fine Tuning DistilBERT



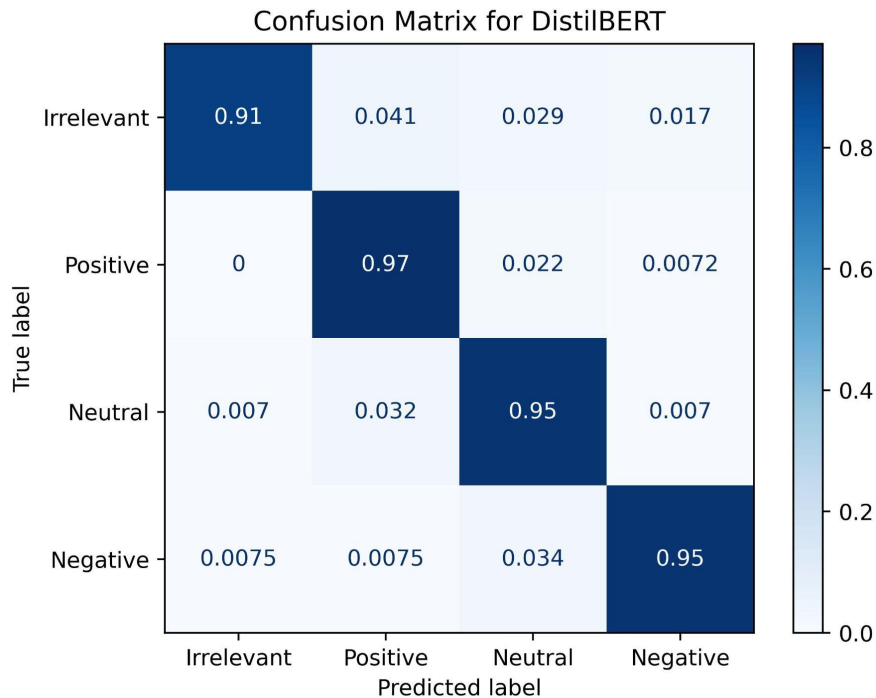
Fine Tuning DistilBERT



Results - DistilBERT

Test Set

MCC	0.934
Avg. Precision	0.95
Avg. Recall	0.95
Avg. F1	0.95
Accuracy	0.95



Conclusions

Model Comparison

Best models were DistilBERT, KNN, and Random Forest

- KNN performed best
- Random Forest and DistilBert almost identical

AdaBoost and Linear SVM performed worse

- Too much complexity for simple division of dataset

Model	Test MCC
KNN	0.954
Random Forest	0.935
DistilBERT	0.934
Linear SVM	0.558
AdaBoost	0.319

Model Comparison

DistilBert took by far the longest to train

AdaBoost notably slower than Random Forest

- Longest training of 20 min vs 4 min

SVM very fast but very inaccurate

KNN quite fast

- 20-25 seconds per fit

Model	Test MCC
KNN	0.954
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Linear SVM	0.558
AdaBoost	0.319

Limitations

Some models could have been run longer

- Resource and time constraints
- AdaBoost, SVM, and Random Forest selected parameters at top of range
 - learning_rate, C, n_estimators respectively

Did not evaluate polynomial or RBF SVMs

- When attempted, took too long per fit to perform a grid search
- Due to translation out of compressed sparse matrix
 - Discovered too late to fix

Future Work

Other models

- Full BERT model or other LLMs
- Different kernels for SVM
- Different classifiers in AdaBoost

Apply to real-world data

- Tracking trends of sentiment

Compare performance on “out-of-domain” data

- Entity classes not in training set

References

Dataset:

<https://www.kaggle.com/datasets/jp797498e/twitter-entity-sentiment-analysis>

Heavy use was made of the examples and tutorials provided on HuggingFace's website in order to write the code needed to train the DistilBERT model:

<https://huggingface.co/docs/transformers/>

References - Papers

BERT

J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, 'BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding', arXiv [cs.CL]. 2019.

<https://arxiv.org/abs/1810.04805>

DistilBERT

V. Sanh, L. Debut, J. Chaumond, and T. Wolf, 'DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter', arXiv [cs.CL]. 2020.

<https://arxiv.org/abs/1910.01108>

Questions

Thank you for listening